



Hoppers Challenge: Carbon Portfolio Diversity

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A sustainable world is one in which our environmental systems and societies flourish, natural resources are able to replenish themselves, and people are able to live healthy and secure lives. The growing complexity and scale of scientific computing workflows and AI training pipelines place huge demands on energy and other natural resources. Sustainability can be overlooked in favour of technological advancements, therefore it is of utmost importance for us to make the effort to consider how we can fit sustainability into our own projects.

The Challenge

Build a carbon credit portfolio that delivers 100,000 of CO₂ e on a budget and survives the unexpected. Try to minimize carbon emissions.

We have used the [UC Berkeley Voluntary Registry Offsets Database](#) with a combination of fake synthetic prices and failure probabilities.

Link to created [DataBase](#) with the fake data created

How will we judge?

Marks will be given for the following:

- Teamwork (how much did everyone contribute)
- Originality
- Theme (How well does your project fit the theme of Sustainability?)
- Presentation Skills
- Project Quality

Extra Resources

<https://pypi.org/project/yfinance/>

<https://climatetrace.org/data>

<https://github.com/ranaroussi/yfinance>

[https://www.eea.europa.eu/en/datahub?](https://www.eea.europa.eu/en/datahub?size=n_10_n&filters%5B0%5D%5Bfield%5D=issued.date&filters%5B0%5D%5Btype%5D=any&filters%5B0%5D%5Bvalues%5D%5B0%5D=All%20time)

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<https://ourworldindata.org/co2-and-greenhouse-gas-emissions#explore-data-on-co2-and-greenhouse-gas-emissions>

Good luck!